

IPv6

In Depth

FIG_000 · guide v1.0 · 2026 · open reference · 9 sections

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Beyond the comparison table – how IPv6 addresses are actually structured, how devices configure themselves without DHCP, and what changes (and doesn't) when there's finally enough address space for everyone.

```
$ read section_01 --start
```

WHAT'S INSIDE

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4 – 5	SLAAC self-configuration, and Neighbor Discovery Protocol
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How to use this guide: read start to finish for the full picture, or jump to section 2 if you just need address notation clarified. Every diagram is numbered and referenced directly in the surrounding text.

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This guide is designed to be read section by section, with diagrams positioned next to the concepts they illustrate.

Why IPv6: The Address Crisis Recap

1.1 THE NUMBERS

IPv4's 32-bit address space provides roughly 4.3 billion unique addresses – a number that seemed enormous in the 1980s and was fully exhausted at the regional registry level years ago. **IPv6**, standardized in 1998, uses 128-bit addresses instead, providing approximately 340 undecillion (3.4×10^{38}) addresses – enough, famously, to assign a unique address to every grain of sand on Earth many times over.

NAT was the band-aid, not the cure. The main Computer Networks guide and the NAT Deep Dive guide both cover how NAT (and later CGNAT) let IPv4 survive years past its natural address-exhaustion date by sharing addresses. IPv6 was designed to make that sharing unnecessary altogether – though as covered in the NAT guide, Section 1.3, NAT has persisted for reasons beyond pure scarcity.

1.2 WHAT IPV6 ACTUALLY CHANGES

- **Address space** – effectively unlimited for any practical purpose, restoring true end-to-end addressability.
- **Simplified header** – a fixed 40-byte header (versus IPv4's variable length), no header checksum (upper-layer protocols already checksum, making it redundant), enabling faster router processing.
- **Built-in autoconfiguration** – SLAAC (Section 4) lets devices configure their own addresses without a DHCP server, though DHCPv6 still exists for networks that want centralized control.
- **No more NAT requirement** – with abundant address space, every device can have its own globally routable address again, though private deployments still often use NAT66-equivalent techniques for other reasons.

1.3 ADOPTION HAS BEEN SLOW, NOT ABSENT

Despite being over 25 years old, meaningful global IPv6 adoption took a long time to build momentum – largely due to the inertia effect already covered in the NAT guide: IPv4 with NAT kept working well enough that migration pressure stayed low for a long time. That said, adoption has accelerated significantly in recent years, and dual-stack deployment (Section 7) is now common practice at major providers and mobile carriers worldwide.

IPv6 Address Anatomy & Notation

2.1 128 BITS, WRITTEN IN HEXADECIMAL

An IPv6 address is 128 bits, written as eight groups of four hexadecimal digits, separated by colons – for example: `2001:0db8:85a3:0000:0000:8a2e:0370:7334`. Each group represents 16 bits, and hexadecimal is used (rather than IPv4's decimal) simply because it maps much more compactly onto binary – each hex digit represents exactly 4 bits.



Fig 2.1 – A full IPv6 address: eight 16-bit hex groups, conventionally split into a 64-bit network prefix and a 64-bit interface identifier.

2.2 SHORTENING RULES

Two rules make IPv6 addresses shorter to write in practice:

- **Leading zeros in each group may be omitted** – `0db8` becomes `db8`, `0000` becomes `0`.
- **One consecutive run of all-zero groups may be replaced with `::`** – used at most once per address, since using it twice would make the number of zero groups ambiguous.

Applying both rules, the address above shortens to: `2001:db8:85a3::8a2e:370:7334`.

2.3 CIDR NOTATION

Just like IPv4, IPv6 uses CIDR notation to indicate the network prefix length – e.g., `2001:db8:85a3::/64` means the first 64 bits are the network portion. A `/64` is the conventional, near-universal subnet size for a single IPv6 network segment (Section 8 covers this in detail), a striking contrast to IPv4 where subnet sizes are tuned tightly to expected host counts.

Address Types: Unicast, Multicast & Anycast

3.1 UNICAST: ONE-TO-ONE

The standard case: an address identifying exactly one interface. IPv6 unicast addresses come in a few important flavors:

TYPE	PREFIX	SCOPE
Global Unicast	2000::/3	Globally routable – the IPv6 equivalent of a public IPv4 address
Link-Local	fe80::/10	Valid only on the local link; never routed – every IPv6 interface has one automatically
Unique Local	fc00::/7	Private, like RFC 1918 – not globally routable, roughly IPv6's answer to 10.0.0.0/8
Loopback	::1	The single address, analogous to IPv4's 127.0.0.1

Link-local addresses aren't optional. Unlike IPv4 where every interface needs manual or DHCP-assigned configuration to do anything at all, every IPv6 interface automatically self-assigns an fe80::/10 link-local address the moment it comes up – this is what makes protocols like NDP (Section 5) work even before any other address configuration has happened.

3.2 MULTICAST: ONE-TO-MANY

Multicast (ff00::/8) sends a single packet to a defined group of interested receivers – conceptually the same idea covered in the main Computer Networks guide's glossary, just far more central to how IPv6 itself operates. Multicast entirely replaces IPv4's broadcast concept – **IPv6 has no broadcast address at all**; anything that would have been a broadcast in IPv4 (like NDP's neighbor discovery, Section 5) uses a specific multicast group instead, reducing unnecessary interruption to devices that aren't relevant recipients.

3.3 ANYCAST: ONE-TO-NEAREST

An **anycast** address is assigned to multiple interfaces (often on different physical devices, sometimes in different locations entirely), and a packet sent to it is delivered to whichever one is *topologically nearest*, as determined by routing. This is exactly the concept behind global DNS root server deployment and CDN edge routing – IPv6 formalizes it as a first-class address type rather than requiring routing tricks to approximate it, as is common in IPv4.

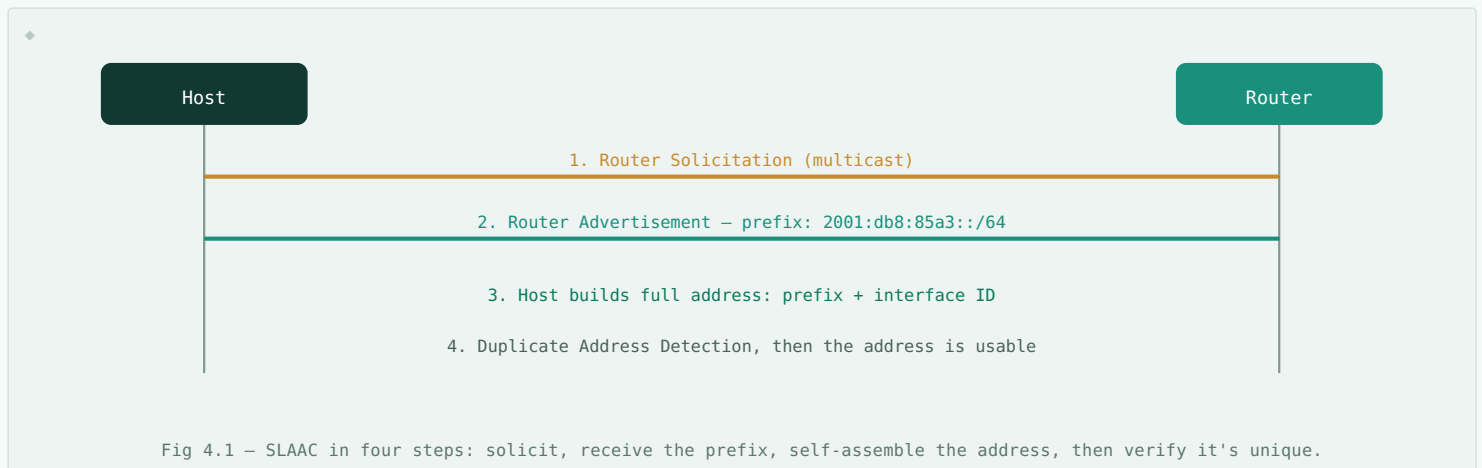
SLAAC: Stateless Address Autoconfiguration

4.1 CONFIGURING YOURSELF, NO DHCP REQUIRED

SLAAC (Stateless Address Autoconfiguration) lets an IPv6 device generate its own fully valid, globally routable address without ever talking to a DHCP server – a genuinely new capability compared to IPv4, where DHCP or manual configuration is essentially mandatory for anything beyond a link-local-equivalent address.

4.2 HOW SLAAC ACTUALLY WORKS

1. The device generates its own link-local address (`fe80::/10`) immediately on startup, using its interface's MAC address or a randomly generated identifier.
2. It sends a **Router Solicitation** message (part of NDP, Section 5) asking any local routers to identify themselves.
3. A router replies with a **Router Advertisement**, containing the network's prefix (typically a /64, Section 2.3) and other configuration parameters.
4. The device combines the advertised prefix with its own interface identifier to construct a full global unicast address – entirely locally, with no central server ever allocating or tracking it.
5. Before using the new address, the device performs **Duplicate Address Detection** – briefly checking via NDP that no other device on the link is already using the same address.



4.3 SLAAC VS DHCPV6

SLAAC handles addressing, but not everything a network might need to configure (DNS servers, for instance, historically required DHCPv6 or router-advertised options). Many networks run SLAAC for address assignment alongside **stateless DHCPv6** purely for supplementary configuration – or **stateful DHCPv6** instead of SLAAC entirely, when centralized address tracking and control matter more than the zero-configuration convenience.

NDP: Neighbor Discovery Protocol

5.1 IPV6'S REPLACEMENT FOR ARP (AND MORE)

NDP (Neighbor Discovery Protocol) is the IPv6 mechanism for discovering other devices on the local link, resolving addresses, and finding routers – consolidating what IPv4 spread across several separate protocols (ARP, ICMP Router Discovery, ICMP Redirects) into one unified, more capable framework built on ICMPv6 messages.

5.2 THE FIVE NDP MESSAGE TYPES

MESSAGE	PURPOSE
Router Solicitation (RS)	Host asks local routers to identify themselves (SLAAC, Section 4)
Router Advertisement (RA)	Router announces itself, its prefix, and configuration parameters
Neighbor Solicitation (NS)	Ask "who has this IPv6 address?" – IPv6's direct replacement for ARP requests
Neighbor Advertisement (NA)	Reply confirming an address-to-MAC mapping – replaces ARP replies
Redirect	Router informs a host of a better next-hop for a specific destination

5.3 WHY CONSOLIDATING INTO NDP MATTERS

Beyond just tidiness, NDP messages use IPv6's built-in security options (like **SEND – Secure Neighbor Discovery**) more consistently than IPv4's patchwork of separate protocols ever could, and – since NDP relies on multicast rather than IPv4's ARP broadcasts – every device on the link isn't interrupted by every single address resolution, only the ones actually relevant to it (tying back to Section 3.2's point about IPv6 having no broadcast address at all).

Address Resolution, renamed and rebuilt: if you're comfortable with how ARP works from the main Computer Networks guide, Neighbor Solicitation/Advertisement is functionally the same idea – "what's the MAC address for this IP?" – just carried over ICMPv6 multicast instead of an Ethernet broadcast frame.

IPv6 vs IPv4: Key Differences

A deeper comparison than the summary table in the main Computer Networks guide – the specific mechanisms behind each difference, covered across this guide's earlier sections.

ASPECT	IPv4	IPv6
Address length	32 bits (~4.3 billion addresses)	128 bits (~340 undecillion addresses)
Address configuration	Manual or DHCP required	SLAAC (Section 4) – self-configuring, or DHCPv6
Address resolution	ARP (broadcast-based)	NDP (Section 5) – multicast-based
Broadcast	Exists – interrupts every device on the segment	Doesn't exist – replaced entirely by multicast (Section 3.2)
Header	Variable length, includes a checksum	Fixed 40 bytes, no checksum – faster router processing
Fragmentation	Can happen at any router along the path	Only at the source host – routers never fragment in transit
Built-in security	IPsec optional, bolted on	IPsec support designed in from the start (though not mandatory in practice)
NAT necessity	Essentially required at internet scale	Not required – abundant address space allows direct end-to-end addressing

The fragmentation change has real consequences. Because only the originating host can fragment an IPv6 packet, **Path MTU Discovery** – the process of finding the smallest MTU along a path – becomes essential rather than optional. If a router along the way can't forward an oversized packet, it drops it and sends back an ICMPv6 "Packet Too Big" message rather than fragmenting it itself, and the sending host is expected to adjust and resend at the correct size.

Transition Mechanisms

7.1 THE COEXISTENCE PROBLEM

IPv4 and IPv6 are not directly interoperable – a pure IPv6-only host cannot natively talk to a pure IPv4-only host without some form of translation or tunneling in between. Since the internet can't switch over instantly, several **transition mechanisms** exist to let both protocols coexist during the (very extended) migration period.

7.2 DUAL-STACK: RUN BOTH, SIMULTANEOUSLY

The most common and straightforward approach: a device runs both IPv4 and IPv6 stacks at once, with both addresses assigned to the same interface. Applications and the operating system negotiate which protocol to use per-connection (commonly via a mechanism like Happy Eyeballs, which races both and uses whichever responds first). This is how most modern dual-stack networks and end-user devices actually operate today.

7.3 TUNNELING: IPV6 OVER IPV4 INFRASTRUCTURE

When a device or network only has IPv4 connectivity available but needs to reach IPv6 destinations, **tunneling** mechanisms (like 6to4 or Teredo, largely historical at this point, and more current approaches like 6in4) encapsulate IPv6 packets inside IPv4 packets for transport across IPv4-only infrastructure, unwrapping them back to native IPv6 at the other end – conceptually similar to how a VPN tunnel (VPN Protocols guide) wraps one protocol's traffic inside another for transport.

7.4 NAT64 & DNS64: TRANSLATING BETWEEN PROTOCOLS

For an IPv6-only client to reach an IPv4-only destination (common on mobile networks that are IPv6-only internally), **NAT64** performs actual protocol translation – not just address rewriting like traditional NAT, but converting the packet itself between IPv4 and IPv6 formats. **DNS64** pairs with it, synthesizing an IPv6 address for IPv4-only destinations that lack a native AAAA DNS record, so the client's request can be routed to the NAT64 gateway transparently.

MECHANISM	SOLVES	COMPLEXITY
Dual-Stack	Native support for both protocols on one device	Low – the default modern approach
Tunneling (6in4, etc.)	IPv6 connectivity over IPv4-only infrastructure	Medium – requires tunnel endpoints
NAT64 / DNS64	IPv6-only clients reaching IPv4-only destinations	High – actual protocol translation

Worked Example: Reading & Subnetting an Address

Given the address `2001:0db8:85a3:0042:0000:0000:0000:1a2b/64`, let's break it down using the concepts from earlier sections.

8.1 SHORTENING IT (SECTION 2.2)

Removing leading zeros and collapsing the longest run of all-zero groups: `2001:db8:85a3:42::1a2b`.

8.2 IDENTIFYING THE PREFIX & INTERFACE ID

COMPONENT	VALUE	MEANING
Network Prefix (/64)	<code>2001:db8:85a3:0042::</code>	Identifies the specific subnet – the first 4 groups
Interface Identifier	<code>::1a2b</code>	Identifies this specific device within that subnet – the last 4 groups

8.3 SUBNETTING A /48 ALLOCATION

Organizations are commonly assigned a `/48` block (65,536 possible /64 subnets) or a `/56` block (256 possible /64 subnets) by their provider. Subnetting in IPv6 almost always means slicing up the space *between* the assigned prefix and the standard /64 boundary – the interface identifier portion is conventionally left alone at exactly 64 bits, unlike IPv4 where the split between network and host bits is tuned per-subnet based on how many hosts are actually needed.

SUBNET	ADDRESS RANGE
1	<code>2001:db8:85a3:0000::/64</code>
2	<code>2001:db8:85a3:0001::/64</code>
3	<code>2001:db8:85a3:0002::/64</code>
...	up to <code>2001:db8:85a3:ffff::/64</code> (65,536 total, from a /48)

Why always /64? SLAAC (Section 4) is specifically designed around a 64-bit interface identifier – deviating from a /64 subnet size breaks SLAAC's address self-assembly on that segment. This is why, unlike IPv4's tightly-tuned host-count-based subnetting, IPv6 subnet planning is almost always just "how many /64s do I need," not "how many hosts per subnet."

Quick Reference & Glossary

TERM	DEFINITION
IPv6	128-bit addressing protocol succeeding IPv4, providing a vastly larger address space.
Global Unicast	A globally routable IPv6 address, the equivalent of a public IPv4 address.
Link-Local	An automatically-assigned address (fe80::/10) valid only on the local link, never routed.
Unique Local	A private IPv6 address range (fc00::/7), IPv6's rough equivalent of RFC 1918 space.
Multicast	One-to-many delivery; entirely replaces IPv4's broadcast concept in IPv6.
Anycast	An address shared by multiple interfaces, delivered to the topologically nearest one.
SLAAC	Stateless Address Autoconfiguration – lets a device self-assign an address without DHCP.
NDP	Neighbor Discovery Protocol – IPv6's consolidated replacement for ARP and related IPv4 protocols.
Router Advertisement	A message a router sends announcing its prefix and configuration parameters.
Duplicate Address Detection	The process of verifying a newly formed address isn't already in use before adopting it.
Dual-Stack	Running IPv4 and IPv6 simultaneously on the same device or network.
NAT64	A mechanism translating between IPv6 and IPv4 packets, not just rewriting addresses.
Path MTU Discovery	The process of finding the smallest MTU along a path, essential since IPv6 routers never fragment.

End of guide. IPv6 isn't just "IPv4 with more addresses" – SLAAC, NDP, and the removal of broadcast entirely change how devices find each other on a network. The mental models transfer from IPv4, but almost none of the specific mechanisms carry over unchanged.